

ServiceNow Script-Controlled ACL

Project Documentation Format

1. Introduction

- Project Title: Script-Controlled ACL – Restrict Record Access Based on Field Value
- Team Members: To be provided; roles include ServiceNow administrator, ACL developer, tester, and project owner.

2. Project Overview

- Purpose: Implement record-level security so access to Institution Details records is determined by the Branch field and the user's assigned roles.
- Features: EEE-only read access for bb1 users, administrator read override, and separate Create, Write, and Delete controls using bb2, bb3, and bb4.

3. Architecture

- Platform: ServiceNow instance with the System Security, User Administration, System Definition, and table/list/form interfaces.
- Security layer: Record ACLs on u_institution_details, including a scripted Read ACL and operation-specific role checks.
- Data model: Custom table u_institution_details with Student Roll Number (auto-number), Student Name (User reference), Faculty Name (User reference), Branch (ECE/EEE/CSE choice), Email, Phone Number, and Description.
- Decision logic: administrators are allowed; bb1 is required for the EEE read path; bb2, bb3, and bb4 govern Create, Write, and Delete.

4. Setup Instructions

- Prerequisites: Authorized ServiceNow admin and security_admin access; test user EEE User; roles bb1, bb2, bb3, bb4; and a non-privileged test user.
- Installation/configuration: create the custom table and fields, create roles, assign roles, create ECE/EEE/CSE records, elevate security_admin, and configure the four ACLs. Do not store passwords in this document.

5. Configuration Structure

- User Administration: EEE User and test users.
- Roles: bb1 (EEE read), bb2 (Create), bb3

(Write), bb4 (Delete). • System Definition: u_institution_details table and fields.

• System Security: record ACLs for read, create, write, and delete. • Validation views: u_institution_details list, record form, and Service Portal view where enabled.

6. Running the Application

- Open the authorized ServiceNow instance.
- Navigate to the relevant table/list through the Application Navigator or open the configured u_institution_details list.
- Impersonate admin, EEE User, and users with different role combinations to validate access decisions.
- For administration: elevate security_admin before editing ACLs.
- For testing: use disposable records for Create, Write, and Delete tests and restore test data after execution.

7. ACL and Interface Documentation

- Read ACL: record / read / u_institution_details; active and advanced; bb1 role; Branch is EEE data condition; script allows admin and bb1 users and denies others.
- Create ACL: record / create / u_institution_details; bb2 required.
- Write ACL: record / write / u_institution_details; bb3 required.

Delete ACL: record / delete / u_institution_details; bb4 required.

• No custom REST API endpoints are defined in the supplied project file;

operations are performed through ServiceNow UI and ACL evaluation.

8. Authentication and Authorization

• Authentication uses ServiceNow user accounts and the instance's configured login method. • Authorization uses ServiceNow roles and record ACLs. The Read script checks `gs.hasRole('admin')` and `gs.hasRole('bb1')`; operation-specific ACLs require bb2, bb3, or bb4. Passwords, tokens, and session secrets are not documented here.

9. User Interface

• List view: validates record visibility by Branch and role. • Form view: validates opening and editing records. • New and Delete controls: validate Create and Delete ACLs. • Service Portal: repeat checks where the table is exposed through a configured portal widget.

10. Testing

• **Functional testing covers positive and negative Read, Create, Write, and Delete scenarios for admin, EEE User, and unauthorized users.** • **Performance testing measures list/form response, filtered-list behavior, controlled concurrency, and sustained-session stability.** • **Record actual results, evidence, response times, and defects during execution; no results are fabricated in this document.**

11. Screenshots or Demo

• Required evidence: user/role records, table and field definitions, each ACL configuration,

EEE-only list view, admin all-branch view, denied access, New/Edit/Delete behavior, and

performance results. Screenshots and instance/demo links: To be added after execution.

12. Known Issues

• The ServiceNow instance URL, team details, credentials, performance thresholds, screenshots, and executed test results were not included in the supplied project file and remain to be completed. • Test data should be disposable for destructive ACL tests. • Service Portal validation depends on the table being exposed in that interface.

13. Future Enhancements

• Add automated ServiceNow ATF coverage for every role/Branch/operation combination. • Add audit dashboards and scheduled ACL regression tests. • Expand policy rules to support additional branches and departments. • Add deployment controls, code review, and CI/CD validation for ACL changes.